Huawei Access Node MALL-ACC-03 Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) Resolution SOP

Node: MALL-ACC-03 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_1503

Description

The Huawei Access Node MALL-ACC-03 Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) is a common issue that can occur in the Access role due to the inherent vulnerabilities in the SFP/XFP transceivers. The SFP/XFP transceivers are used to transmit and receive optical signals over long distances. However, they can degrade over time due to various factors such as fiber cuts, Rx/Tx power loss, and environmental conditions. In the Huawei Access Node MALL-ACC-03, the SFP/XFP transceivers are integrated into the device and are used to connect the access node to the core network. When an SFP/XFP transceiver degrades or fails, it can cause a loss of signal, leading to a minor alarm (ALARM_1503). This SOP outlines the steps to diagnose and resolve the issue.

Pre-requisite Checks

- Ensure that the Malleswaram Node is accessible and that the remote hands are dispatched to the site.

- Verify that the redundant link status is checked and that the link is operational.

- Ensure that the necessary tools and equipment are available to perform the diagnostic and resolution tasks.

Potential Root Causes

Cause	Probability
SFP/XFP transceiver degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display transceiver info, Expected Output: Transceiver Info: Module ID: 0 Type: SFP-XFP Status: Degraded Rx Power: -20 dBm Tx Power: -15 dBm, Description: This command displays the transceiver information, including the module ID, type, status, Rx power, and Tx power. This command helps to identify the degraded SFP/XFP transceiver and its location.

Command: display fiber info, Expected Output: Fiber Info: Fiber ID: 1 Type: SMF Length: 10 km Attenuation: 3 dB, Description: This command
```

```
displays the fiber information, including the fiber ID, type, length, and attenuation. This command helps to identify the fiber cut or degradation and its location.
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Resolution Steps

Step 1: Step: Replace the degraded SFP/XFP transceiver, Command: replace transceiver 0, Description: This step involves replacing the degraded SFP/XFP transceiver with a new one. This is done to restore the signal and resolve the issue.

Step 2: Step: Verify the fiber integrity, Command: verify fiber integrity 1, Description: This step involves verifying the fiber integrity and checking for any cuts or degradation. This is done to ensure that the fiber is operational and not causing the issue.

Post-Check Verification

```
Command: display transceiver info, Expected Output: Transceiver Info: Module ID: 0 Type: SFP-XFP Status: Normal Rx Power: -15 dBm Tx Power: -10 dBm, Description: This command displays the transceiver information after the resolution steps. The expected output is a normal status and the desired Rx and Tx power levels.

Command: display fiber info, Expected Output: Fiber Info: Fiber ID: 1 Type: SMF Length: 10 km Attenuation: 3 dB, Description: This command displays the fiber information after the resolution steps. The expected output is a normal fiber status and the desired attenuation level.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: restore transceiver 0, Description: This step involves restoring the original SFP/XFP transceiver that was replaced. This is done in case the new transceiver does not resolve the issue or causes further problems.

- Step: Verify the fiber integrity, Command: verify fiber integrity 1, Description: This step involves verifying the fiber integrity again after the rollback. This is done to ensure that the fiber is operational and not causing the issue.